4、 $u = v + w$ ，方程非，边界齐

$$\frac{\partial^2 v}{\partial t^2} - a^2\frac{\partial^2 v}{\partial x^2} = f(x,t), \quad 0<x<l,\ t>0$$
$$v(x,t)|_{x=0} = 0, \quad v(x,t)|_{x=l} = 0$$

$$\frac{\partial^2 w}{\partial t^2} - a^2\frac{\partial^2 w}{\partial x^2} = 0, \quad 0<x<l,\ t>0$$
$$w|_{x=0} = 0, \quad w|_{x=l} = 0, \quad t \ge 0$$
$$w|_{t=0} = -v|_{t=0}, \quad \frac{\partial w}{\partial t}|_{t=0} = -\frac{\partial v}{\partial t}|_{t=0}, \quad 0 \le x \le l$$

构造 v

5、 f 复杂，令 $f = \sum_{n=1}^{\infty} g_n(t) X_n(x)$

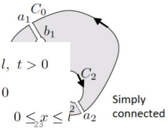
$X_n(x)$ 为 $X_n''(x) + \lambda_n X_n(x) = 0$ 特征函数

（利用正交性得展开式 $g_n(t)$ ）

$$T_n''(t) + \lambda_n a^2 T_n(t) = g_n(t)$$

若 $u|_{t=0}$ ， $\frac{\partial u}{\partial t}|_{t=0} = 0$

则 $T_n(t) = \frac{l}{n\pi a}\int_0^t g_n(\tau) \sin\frac{n\pi}{l}a(t-\tau)\,d\tau$



equivalent to

$$\oint_{C_0} f(z)dz = \oint_{C_1^-} f(z)dz + \oint_{C_2^-} f(z)dz$$

Here C_i^- keeps the same direction as C_0 (i.e. clockwise or anti-clockwise)

$$\oint_C z^n dz = \oint_{|z|=\varepsilon} z^n dz = \int_0^{2\pi} \varepsilon^{n+1} e^{i(n+1)\theta} i d\theta = \begin{cases} 2\pi i, & n=-1; \\ 0, & n=-2,-3,-4,\dots; \end{cases}$$

计算 Taylor Series

① 裂项

② 级数相乘

③ 求导 $\frac{1}{(1-z)^2}$

④ 定义 $a_n = \frac{f^{(n)}(a)}{n!}$

求 u、v

$$dv(x,y) = \frac{\partial v}{\partial x}dx + \frac{\partial v}{\partial y}dy \qquad \text{Full derivative}$$

由 CR

$$v(x,y) = \int dv(x,y) = \int -\frac{\partial u}{\partial y}dx + \frac{\partial u}{\partial x}dy$$

解析性质

① Laplace's equation

$$\Delta u = \nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$
$$\Delta v = \nabla^2 v = \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$$

② Taylor series 收敛

③ u, v 满足 Laplace 方程

则必为某解析函数 $f(z)$

实或虚部

解析例子

① 简单函数

② 双曲函数

③ $f(g(z))$ ，f、g 解析

线积分

$$\int_\gamma f(z)\,dz = \int_\gamma (u+iv)(dx+idy) = \int_C (udx - vdy) + i\int_C (vdx + udy)$$

$$\int_\gamma f(z)\,dz := \int_a^b f(\gamma(t))\gamma'(t)\,dt.$$

Cauchy's Theorem

$\oint_C f(z)dz = 0$ ，$f(z)$ 在 C 内解析

Laurent series

$$f(z) = a_0 + a_1(z-z_0) + a_2(z-z_0)^2 + \cdots + \frac{b_1}{z-z_0} + \frac{b_2}{(z-z_0)^2} + \cdots$$

zero point

$f^{(m-1)}(a) = 0 \quad f^{(m)}(a) \ne 0 \quad$ m-th order

singular point.

Definition of poles. If only a finite number of the coefficients b_n are nonzero we say z_0 is a finite pole of f. In this case, if $b_k \ne 0$ and $b_n = 0$ for all $n > k$ then we say z_0 is a pole of order k.

order 1 — simple pole

无限 — essential 本性奇点.

$b_n = 0$ — removable 可去

留数定理

$$(5.2) \qquad \oint_C f(z)\,dz = 2\pi i \cdot \text{sum of the residues of } f(z) \text{ inside } C,$$

where the integral around C is in the counterclockwise direction.

方法：

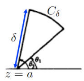
$$R(z_0) = \lim_{z\to z_0}(z-z_0)f(z) \qquad \text{when } z_0 \text{ is a simple pole.}$$

$$a_{-1} = \frac{1}{(m_0-1)!}\frac{d^{m_0-1}}{dz^{m_0-1}}(z-b)^{m_0}f(z)|_{z=b} \text{ with } m_0 \ge m$$

Small arc lemma (小圆弧引理)

If $f(z)$ is continuous in a small region around $z=a$, and satisfies the following relation: when $\theta_1 \le \arg(z-a) \le \theta_2$, $|z-a| \to 0$, $(z-a)f(z)$ uniformly approaches k. Then we have

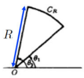
$$\lim_{\delta\to 0}\int_{C_\delta} f(z)dz = ik(\theta_2 - \theta_1)$$



Great arc lemma (大圆弧引理)

If $f(z)$ is continuous in a region around $z=\infty$, and satisfies the following relation: when $\theta_1 \le \arg(z) \le \theta_2$, $z\to\infty$, $zf(z)$ uniformly approaches K. Then we have

$$\lim_{R\to\infty}\int_{C_R} f(z)dz = iK(\theta_2-\theta_1)$$



二阶 ODE

① y miss

$y' = p \quad y'' = p'$

② x miss

$y' = p \quad y'' = p\frac{dp}{dy}$

③ $y'' + f(y) = 0$

$\Rightarrow y'dy' + f(y)dy = 0$ 积分

④ $a_2 x^2 \frac{d^2y}{dx^2} + a_1 x \frac{dy}{dx} + a_0 x = f(x)$

令 $x = e^z$

$x\frac{dy}{dx} = \frac{dy}{dz}, \quad x^2\frac{d^2y}{dx^2} = \frac{d^2y}{dz^2} - \frac{dy}{dz}$

$\Rightarrow a_2\frac{d^2y}{dz^2} + (a_1 - a_2)\frac{dy}{dz} + a_0 y = f(e^z)$

⑤ 已知一解 u

设 $y = u\cdot v$ 代入

求另一解

PDE

① 方程齐次 多项式齐次

$L(D_x, D_y) = A_0(D_x - a_1 D_y)(D_x - a_2 D_y)\cdots$

$u = \phi_1(y + a_1 x) + \phi_2(y + a_2 x) + \cdots \phi_n$

$D_x D_y = D_y D_x$

$A_0 a^n + A_1 a^{n-1} + \cdots A_n = 0$

$a_1, a_2, \cdots a_n$

$(D_x - a_1 D_y)^n u = 0$

$u = x^{n-1}\phi_1(y + a_1 x) + x^{n-2}\phi_2(y + a_1 x) + \cdots + \phi_n(y + a_1 x)$

② $(D_x - aD_y - b)y = 0$

$u = e^{bx}\phi(y + ax)$

$(D_x - aD_y - b)^n y = 0$

$u = x^{n-1}e^{bx}\phi_1(y + ax) + x^{n-2}e^{bx}\phi_2(y + ax) + \cdots + e^{bx}\phi_n(y + ax)$

③ 方程非齐次

$L(D_x, D_y)u = f(x, y)$

1. $f(x,y) = e^{ax+by}$

$u_0 = \frac{1}{L(D_x, D_y)}e^{ax+by} = \frac{1}{L(a,b)}e^{ax+by}$

But if $L(a,b) = 0$, without losing generality, we can assume

$\hat{L}(\hat{D}_x, \hat{D}_y) = b\hat{D}_x - a\hat{D}_y, \quad [L(a,b) = b\cdot a - a\cdot b = 0]$

$(b\hat{D}_x - a\hat{D}_y)u = e^{ax+by}$

Plug in

we assume the special solution as $u_0(x,y) = f(x,y)e^{ax+by}$

$(b\hat{D}_x - a\hat{D}_y)f(x,y) = 1$ Let's further assume $f(x,y) = \alpha x + \beta y + \gamma$ ➡ $b\alpha - a\beta = 1$

- If we take $\beta = \gamma = 0, \ \alpha = 1/b \quad u_0 = (x/b)\cdot e^{ax+by}$
- If we take $\alpha = \gamma = 0, \ \beta = -1/a \quad u_0 = (-y/a)\cdot e^{ax+by}$

(2) If $f(x,y) = e^{i(ax+by)}$, then

$$\frac{1}{\hat{L}(\hat{D}_x, \hat{D}_y)}e^{i(ax+by)} = \frac{1}{L(ia, ib)}e^{i(ax+by)}$$

When *a* & *b* are real numbers, and when the coefficients of the PDE are real numbers

$$\frac{1}{\hat{L}(\hat{D}_x, \hat{D}_y)}\sin(ax+by) = \mathrm{Im}\left[\frac{1}{L(ia, ib)}e^{i(ax+by)}\right]$$

$$\frac{1}{\hat{L}(\hat{D}_x, \hat{D}_y)}\cos(ax+by) = \mathrm{Re}\left[\frac{1}{L(ia, ib)}e^{i(ax+by)}\right]$$

(3) If $f(x,y) = e^{ax+by}g(x,y)$, then

$\hat{L}(\hat{D}_x, \hat{D}_y)$

$$\frac{1}{\hat{L}(\hat{D}_x, \hat{D}_y)}e^{ax+by}g(x,y) = e^{ax+by}\frac{1}{\hat{L}(\hat{D}_x + a, \hat{D}_y + b)}g(x,y)$$

Easy to verify the above, because

$$\left.\begin{aligned}\hat{D}_x\left[e^{ax+by}g(x,y)\right] &= e^{ax+by}(\hat{D}_x + a)g(x,y)\\ \hat{D}_y\left[e^{ax+by}g(x,y)\right] &= e^{ax+by}(\hat{D}_y + b)g(x,y)\end{aligned}\right\} \hat{L}(\hat{D}_x, \hat{D}_y)\left[e^{ax+by}g(x,y)\right] = e^{ax+by}\hat{L}(\hat{D}_x + a, \hat{D}_y + b)g(x,y)$$

$$\boxed{\begin{aligned}&\hat{L}(\hat{D}_x, \hat{D}_y)\left[e^{ax+by}\frac{1}{\hat{L}(\hat{D}_x + a, \hat{D}_y + b)}g(x,y)\right]\\ &= e^{ax+by}\hat{L}(\hat{D}_x + a, \hat{D}_y + b)\left[\frac{1}{\hat{L}(\hat{D}_x + a, \hat{D}_y + b)}g(x,y)\right] = e^{ax+by}g(x,y)\end{aligned}}$$

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(4) If $f(x,y) = x^m y^n$, we can expand the operator $\frac{1}{\hat{L}(\hat{D}_x, \hat{D}_y)}$ as a power series of $\hat{D}_x, \ \hat{D}_y$, and then find one special solution.

$$u_0 = \frac{12}{\hat{D}_x^2 - 2\hat{D}_x\hat{D}_y + \hat{D}_y^2}xy = \frac{12}{\hat{D}_x^2}\left(1 - \frac{\hat{D}_y}{\hat{D}_x}\right)^{-2}xy$$

$$= \frac{12}{\hat{D}_x^2}\left(1 + 2\frac{\hat{D}_y}{\hat{D}_x} + \ldots\right)xy = \frac{12}{\hat{D}_x^2}\left(xy + \frac{2}{\hat{D}_x}x\right)$$

$$= 12\left(y\frac{1}{\hat{D}_x^2}x + \frac{2}{\hat{D}_x^3}x\right) = 12\left(\frac{1}{6}x^3 y + \frac{1}{12}x^4\right)$$

$D_y^n xy = 0$

$$\frac{1}{(D_x - aD_y)(D_x - bD_y)}(x^m y^m) = \frac{1}{D_x - aD_y}\left(\frac{1}{D_x - bD_y}x^m y^m\right)$$

分步操作

(5) If $f(x,y) = f(ax+by)$, and if $\hat{L}(\hat{D}_x, \hat{D}_y)$ is homogeneous about $\hat{D}_x, \ \hat{D}_y$ (of order n), then

$\hat{D}_x^r g(ax+by) = a^r g^{(r)}(ax+by), \quad \hat{D}_y^s g(ax+by) = b^s g^{(s)}(ax+by)$

$\hat{L}(\hat{D}_x, \hat{D}_y)g(ax+by) = L(a,b)g^{(n)}(ax+by)$

When $L(a,b) \neq 0$, we have

Easy to verify

Multiply by $\hat{L}(\hat{D}_x, \hat{D}_y)$

$$\frac{1}{\hat{L}(\hat{D}_x, \hat{D}_y)}g^{(n)}(ax+by) = \frac{1}{L(a,b)}g(ax+by)$$

积分n次

分离变量法

$\frac{\partial^2 u}{\partial t^2} - a^2\frac{\partial^2 u}{\partial x^2} = 0$

$u_{x=0} = 0, \quad u_{x=l} = 0$

① 方程齐次，边界条件齐次

1. $u = X(x)T(t)$ 代入方程

$\frac{1}{a^2}\frac{T''(t)}{T(t)} = \frac{X''(x)}{X(x)} = -\lambda$

讨论 $\lambda = 0$

$$\begin{cases} T''(t) + \lambda a^2 T(t) = 0 \\ X''(x) + \lambda X(x) = 0 \end{cases}$$

$u = \sum_{n=0}^{\infty}\left(C_n \sin\frac{n\pi}{l}at + D_n\cos\frac{n\pi}{l}at\right)\sin\frac{n\pi}{l}x$

$\int_0^l \sin^2\frac{n\pi}{l}x \, dx = \frac{l}{2}$

$\lambda_n \neq \lambda_m \quad \int_0^l X_n(x)X_m(x)dx = 0$

2. $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad u|_{x=0} = 0, \ \frac{\partial u}{\partial x}\Big|_{x=a} = 0$

$u = \sum_{n=0}^{\infty}\left(C_n \sinh\frac{2n+1}{2a}\pi y + D_n\cosh\frac{2n+1}{2a}\pi y\right)\cdot\sin\frac{2n+1}{2a}\pi x$

3. $\frac{\partial u}{\partial t} - k\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) = 0$

$\frac{\partial u}{\partial x}\Big|_{x=0} = 0, \ \frac{\partial u}{\partial x}\Big|_{x=a} = 0$

$\frac{\partial u}{\partial y}\Big|_{y=0} = 0, \ \frac{\partial u}{\partial y}\Big|_{y=b} = 0$

$$\begin{cases} X''(x) + \mu X(x) = 0 \\ Y''(y) + \nu Y(y) = 0 \\ T'(t) + \lambda k T(t) = 0 \end{cases}$$

$\mu + \nu = \lambda$ 令 $u = XYT$

$0 < x < a, \ 0 < y < b$

讨论 $\mu, \nu, \lambda = 0$

$\mu_n = \left(\frac{n\pi}{a}\right)^2 \ n = 0,1,2, \ X_n(x) = \cos\frac{n\pi}{a}x$